Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

RP 17: Density

Key concept — density

Density is the mass packed into each unit of volume of a material. Recall this equation: it is not on the equation sheet.

$$\rho = \frac{m}{V} \qquad \text{density} = \frac{\text{mass}}{\text{volume}}$$

- **Rearranged:** $m = \rho V$ and $V = \frac{m}{\rho}$.
- **In the lab:** mass in g, volume in cm^3, so density comes out in g/cm^3.
- **In the exam (SI units):** mass in kg, volume in m^3, density in kg/m^3.
- **To convert,** multiply by 1000: water is $1.00\,\text{g/cm}^3 = 1000\,\text{kg/m}^3$.

Density belongs to the material, not the object. A small and a large block of aluminium have the same density: the large one has more mass and more volume, in proportion.

How do you find the density of a regular solid, an irregular solid and a liquid?

You will measure four sets of solids and some water. Each part has its own table with one column for each object. **Fill in a column for every object you measure.**

Before you start, find the resolution of each instrument on your bench. The **resolution** is the smallest change an instrument can show: the smallest step on its scale, or the last digit on its display.

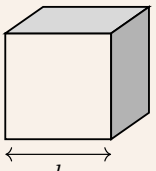
Apparatus	What you use it for	Resolution of yours
Ruler	Sides of the cubes and the cuboids	*1 mm (0.1 cm)*
Vernier callipers	Diameter and length of the small cylinders	*0.1 mm, or as marked*
Balance	Mass of every object, and of the water	*1 g*
Eureka can	Pushing out a volume of water equal to the rock's volume	—
Measuring cylinder	Volume of the water: from the eureka can, and in Part 5	*e.g. 1* cm^3

Safety

- **Wash your hands after handling the lead cube.** Lead is toxic.
- Wipe up spilt water straight away: a wet floor is slippery, and water damages the balances.
- Stand glass measuring cylinders in the middle of the bench.

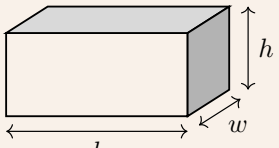
Regular solids — volume from measured sides

Volumes of the three shapes you will measure



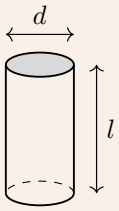
cube

$V = l \times l \times l$



cuboid

$V = l \times w \times h$



cylinder

$V = \pi r^2 l$, with $r = \frac{d}{2}$

Method for every regular solid

1. Switch the balance on with nothing on the pan and press zero. **Check:** it reads 0 g.
2. Place the object on the pan and record its mass.
3. Measure each side. Line the ruler's zero mark up with one edge, and read the other edge with your eye straight above it. Record lengths in cm.
4. Calculate the volume with the formula for its shape, then the density, $\rho = m/V$, to **2 significant figures**.

Part 1 — the density cubes

All twelve cubes come from one set and have the same size. Measure one side of each cube, then use $V = l^3$.

	Brass	Copper	Aluminium	Iron	Zinc	Tin
Mass m / g	*68*	*71*	*22*	*63*	*57*	*58*
Side l / cm	*2.0*	*2.0*	*2.0*	*2.0*	*2.0*	*2.0*
Volume V / cm^3	*8.0*	*8.0*	*8.0*	*8.0*	*8.0*	*8.0*
Density ρ / g/cm^3	*8.5*	*8.9*	*2.8*	*7.9*	*7.1*	*7.3*

	Lead	Acrylic	Polythene	Cork	Hardwood	Softwood
Mass m / g	*90*	*9*	*7*	*2*	*6*	*4*
Side l / cm	*2.0*	*2.0*	*2.0*	*2.0*	*2.0*	*2.0*
Volume V / cm^3	*8.0*	*8.0*	*8.0*	*8.0*	*8.0*	*8.0*
Density ρ / g/cm^3	*11*	*1.1*	*0.88*	*0.25*	*0.75*	*0.50*

Expected for 20 mm cubes on a 1 g balance. Cork and the woods are least certain: 2 g to the nearest gram is ±50%.

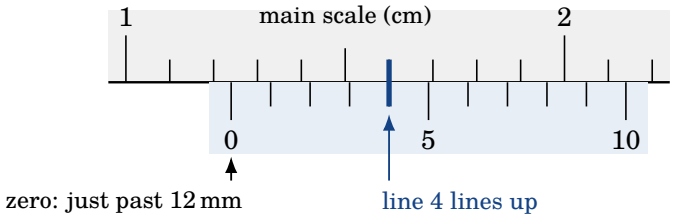
Part 2 — the small cylinders

These are only about 5 mm across, so measure them with the **Vernier callipers**, which read to a fraction of a millimetre.

Reading Vernier callipers

1. Close the jaws gently on the cylinder.
2. **Whole millimetres:** read the main scale at the last line *before* the sliding scale's zero.
3. **The decimal:** find the one sliding-scale line that lines up exactly with a main-scale line. Its number is the decimal part.
4. Add the two. Divide by 10 for cm.

If your callipers have more lines on the sliding scale, the method is the same.



Reading: $12\text{ mm} + 0.4\text{ mm} = 12.4\text{ mm} = 1.24\text{ cm}$

Measure the **diameter** d and the **length** l with the callipers. Find the mass on the balance. Then $r = d/2$, $V = \pi r^2 l$, and $\rho = m/V$.

Material	*brass*	*aluminium*			
Mass m / g	*7*	*2*			
Diameter d / cm	*0.50*	*0.50*			
Length l / cm	*4.00*	*4.00*			
Radius r / cm	*0.25*	*0.25*			
Volume V / cm^3	*0.79*	*0.79*			
Density ρ / g/cm^3	*8.9*	*2.5*			

Part 3 — the cuboids

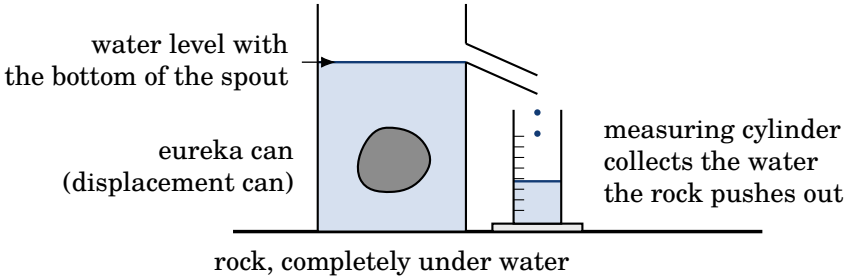
Measure the length, width and height of each with the ruler. $V = l \times w \times h$. Use the spare column for any other cuboid in the set.

	Hardwood	**Softwood**	**Paraffin wax**	**Aluminium**	**Iron**	**Poly-styrene**	
Mass m / g							
Length l / cm							
Width w / cm							
Height h / cm							
Volume V / cm^3							
Density ρ / g/cm^3	*0.6–0.9*	*0.4–0.6*	*0.9*	*2.7*	*7.9*	*1.05 / 0.03*	

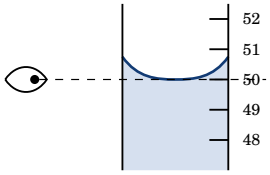
Accepted values. Polystyrene is 1.05 g/cm^3 if solid, about 0.03 if expanded.

Part 4 — the rocks: volume by displacement

A rock has no sides to measure, so you find its volume from the water it pushes out.



1. Find the mass of the dry rock on the balance. Record it.
2. Stand the eureka can by the sink tray and fill it until water runs out of the spout. Let it drip until it stops. **Check:** no drop has fallen for ten seconds.
3. Stand an empty measuring cylinder under the spout.
4. Hold the rock at the water's surface and let it slide in, so the only thing entering the water is the rock. Wait until the spout stops dripping.
5. Read the volume of water in the measuring cylinder, as shown on the right. This is the volume of the rock.
6. Calculate the density, $\rho = m/V$.



The curved surface is the **meniscus**. With your eye level with it, read the **bottom** of the curve. This one reads 50 cm^3.

	Rock A	**Rock B**	**Rock C**
Mass m / g	*135*		
Volume of water pushed out V / cm^3	*50*		
Density ρ / g/cm^3	*2.7*		

Part 5 — water

1. Put the empty, dry measuring cylinder on the balance. Record its mass.
2. Pour in water up to the **50** cm^3 line, reading the bottom of the meniscus with your eye level with it.
3. Put the cylinder back on the balance and record the new mass.
4. Subtract to find the mass of the water, then calculate its density.

Mass of empty cylinder / g	**Mass of cylinder + water / g**	**Mass of water m / g**	**Volume of water V / cm^3**	**Density ρ / g/cm^3**
112	*162*	*50*	50	*1.0*

Before you pack away: wash your hands if you handled the lead cube. Pour the eureka can into the sink, dry the rocks and the cubes, and put every cube back in its box. Wipe the bench and the balance pan dry.

What your results tell you

Material	g/cm^3	kg/m^3	Material	g/cm^3	kg/m^3
Lead	11.3	11 300	Rock (granite, limestone)	about 2.7	about 2700
Copper	8.96	8960	Acrylic	1.18	1180
Brass	8.50	8500	Polystyrene, solid	1.05	1050
Iron	7.87	7870	Water	1.00	1000
Tin	7.29	7290	Polythene	0.92	920
Zinc	7.13	7130	Paraffin wax	0.90	900
Aluminium	2.70	2700	Hardwood (oak)	0.6–0.9	600–900
			Softwood (pine)	0.4–0.6	400–600
			Cork	0.24	240
			Polystyrene, expanded	about 0.03	about 30

Accepted densities. Woods and rocks are natural materials, so each has a range.

1) **Show** one density calculation in full, from any part: the formula first, then your values in place of the symbols, then the answer in a box with its unit.

$$\rho = \frac{m}{V}$$

$$\rho = \frac{22}{8.0}$$

$$\boxed{\rho = 2.8 \text{ g/cm}^3}$$

2) **Compare** the density you measured for **one metal cube** with the accepted value in the table. Suggest one reason why they are not exactly the same.

For example aluminium: measured 2.8, accepted 2.70 g/cm^3. The balance reads only to the nearest gram, so a 21.6 g cube shows 22 g; the side is read to the nearest millimetre, and that small error is cubed in the volume.

3) **Explain** why the twelve cubes have different masses, even though they all have the same volume.

They are made of different materials with different densities. Each material has particles of different mass, packed differently, so the same 8 cm^3 holds a different mass of each.

4) **Compare** your density for the aluminium **cube** with your density for the aluminium **cuboid**. Explain what this shows about density.

They are the same, or very close, although the cuboid is a different size and shape. Density is a property of the material, not of the object: a bigger piece has more mass and more volume in the same proportion.

5) **Convert** the density of your rock into kg/m^3. Show your working.

Multiply by 1000: $2.7 \text{ g/cm}^3 \times 1000 = 2700 \text{ kg/m}^3$.

Evaluation

6) **Explain** why the rocks needed a eureka can, when the cubes did not.

A rock is an irregular shape, so there are no sides to measure and no formula for its volume. The water it pushes out has exactly the same volume as the rock.

7) **Explain** why you used Vernier callipers on the small cylinders but a ruler was good enough for the cubes.

A ruler has a resolution of 1 mm. On a 20 mm cube side that is 5%. On a 5 mm diameter the same 1 mm is 20%, and the radius is squared in $V = \pi r^2 l$, so the volume would be out by about 40%. Vernier callipers read to 0.1 mm or better, which is 2% of 5 mm.

8) **Identify** which measurement limits how accurate your density for a small cylinder can be, and suggest how to improve it.

The mass. A small cylinder weighs only 2–7 g, and the balance reads to the nearest gram, so the mass could be out by up to 50%, far more than the callipers' 2%. Use a balance that reads to 0.01 g. A precise volume cannot rescue an imprecise mass.

9) **Suggest** one change to Part 5 that would make the density of water more accurate.

Use a larger volume, such as 100 cm^3: the 1 g resolution of the balance is then a smaller fraction of the mass. Or repeat the measurement and take a mean, or use a balance with a finer resolution.